

Relational Algebras

But in $\text{\LaTeX}$, 57 years later.

Michael Barr, Typed by Francesco Tognetti

Introduction

All notation not otherwise specified in this paper is taken from the original paper from M. Barr ([Bar69]), or adapted from various pieces of modern literature now considered standard

This text is an attempt to modernize the amazing paper written by M. Barr in 1969, both in terminology, notation and print, as well as added pictures to guide our diagram chasing.

The biggest change is that proofs that in the original paper were presented in algebraic form become more graphic, losing compactness in favor of a clearer diagrammatic visual representation. Outside of this shift, the paper is an attempt of conservative retyping, changing the least possible amount of the original phrasing and explaining, down to paying attention to preserving the original indentation to avoid confusion in whether a new paragraph is supposed to be starting.

Let **Top** denote the category of topological spaces and continuous mappings, **KHaus** the full subcategory whose objects are the compact Hausdorff spaces and **Set** the category of sets.

The obvious functor $U : \mathbf{KHaus} \rightarrow \mathbf{Set}$ has a left adjoint F which is best described by saying that FX is the set of all ultrafilters on X with the hull-kernel topology.

If $\mathbb{B} = (\beta, \eta, \mu)$ is the triple coming from the adjoint pair, the natural functor $\mathbf{KHaus} \rightarrow \mathbf{Set}^{\mathbb{B}}$ is an equivalence.

This associates to a compact Hausdorff space C the pair (UC, c) , where $c : \beta UC \rightarrow UC$ is given by $C(u) = \lim u$ for any ultrafilter $u \in UC$

In an arbitrary topological space V , there is still given a relation $\text{conv} : TUV \rightarrow UV$ which associates to an ultrafilter u the set $\text{conv}(u)$ of all its limits (possibly empty). Moreover it is well known that this convergence relation determines the topology uniquely and that continuity of mappings can also be described by it.

It thus seemed plausible that by a suitable axiomatization of the notion of relational algebra of a triple (or of a theory) it could be shown that **Top** is the category of relational algebras of the theory whose algebras are **KHaus**. That is also the main result of this paper.

Section 1 tabulates a few properties of the category **Rel** of sets and relations.

Section 2 includes the definition of relational pre-algebras and establishes their basic properties.

In Section 3 the main theorem 3.1 about **Top** is proved, and

in Section 4 a few additional examples are given.

Everything here is done for triples over the category **Set**.

Presumably much of this could be done over other categories, at least those in which there is a good notion of what are relations.

Manes conjectures (and is attempting to prove)¹ the analog of 3.1 is true for any varietal category and its compact algebra triple (see [Man69], §5)

In carrying out this work I had several stimulating discussion with Basil Rattray who independently proved (3.1). I would also like to thank the National Research Council of Canada for its support.

¹ He might have proven it in the past 57 years; It is not easy to find.

1 . Relations.

Before talking about relational algebras we will here tabulate some properties of the category **Rel** of sets and relations. If $r : X \rightsquigarrow Y$ is a relation, it has a standard factorization

$$X \xrightarrow{d_r^{-1}} \Gamma_r \xrightarrow{c_r} Y$$

$$x \rightsquigarrow (x, r(x)) \rightsquigarrow r(x)$$

Where $\Gamma_r \subseteq X \times Y$ is the graph of r and the functions d_r, c_r are the restriction to Γ_r of the coordinate projections $d_X : X \times Y \rightarrow X$ on the 'domain' and $c_Y : X \times Y \rightarrow Y$ on the 'codomain'. We write $r^\dagger = r^{-1} := d_r c_r^{-1}$.

$$X \xleftarrow{d_r} \Gamma_r \xleftarrow{c_r^{-1}} Y$$

We call r

epi	if c_r is epi
mono	if c_r is mono
e.d. (everywhere defined)	if d_r is epi
a p.f. (partial function)	if c_r is mono

To define the composite of two relations it is only necessary to define a composite of the form $f^{-1}g$ where f and g are functions, for then we define

$$r \circ s = c_r \circ d_r^{-1} \circ c_s \circ d_s^{-1}.$$

Consider a commutative diagram

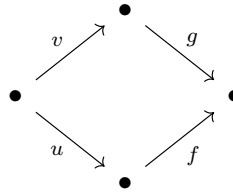


Fig. 1.1

where u, v, f, g are all functions. In general we have only

$$uv^{-1} \subseteq f^{-1}g$$

with equality if 1.1 is a weak pullback. In particular we define $f^{-1}g$ to be uv^{-1} when 1.1 is a pullback.

It is well known that the category **Rel** is a 2-category, the hom-sets being partially ordered by

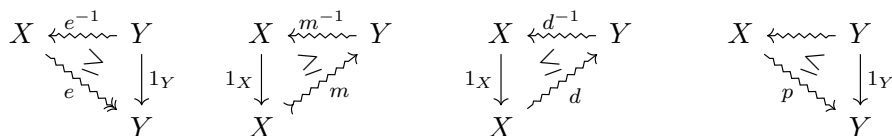
$$r \leq s \text{ in } \mathbf{Rel}(X, Y) \text{ if } \Gamma_r \subseteq \Gamma_s.$$

The following proposition is also well known (see e.g. [Lan61]).

Proposition 1.2. *For any relation $r : X \rightsquigarrow Y$*

- r is epi *iff* $rr^{-1} \supseteq Y$,
- r is mono *iff* $r^{-1}r \subseteq X$,
- r is e.d. *iff* $r^{-1}r \supseteq X$,
- r is a p.f. *iff* $rr^{-1} \subseteq Y$.

We represent these in diagram form as



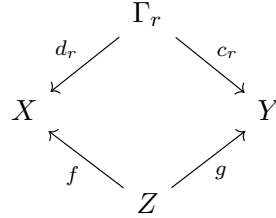
Of course combinations of these can be worked out as well. The most important characterizes r as a function if and only if $r^{-1}r \supseteq X$ and $rr^{-1} \subseteq Y$.

If $T : \mathbf{Set} \rightarrow \mathbf{Set}$ is a functor, it induces a pseudofunctor, also denoted by $T : \mathbf{Rel} \rightarrow \mathbf{Rel}$, given by

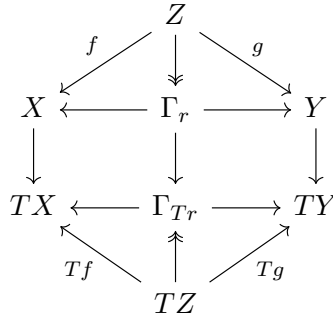
$$T^{\mathbf{Rel}}(c_r d_r^{-1}) = T^{\mathbf{Set}}(c_r) T^{\mathbf{Set}}(d_r)^{-1}.$$

This means that for any r, s composable relations, $T(r \circ s) \subseteq T(r) \circ T(s)$ and that $r \leq s$ implies $T(r) \leq T(s)$. Notice that $T(r^{-1}) = T(r)^{-1}$ under this definition and will be written Tr^{-1} .

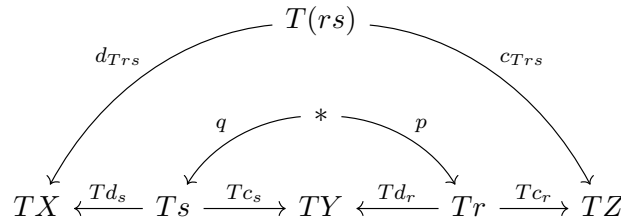
If we have a diagram



then $gf^{-1} \subseteq r$ if and only if there's a mapping $Z \rightarrow \Gamma_r$ making both triangles commute, and $gf^{-1} = r$ if that mapping is epi. In that case, since endofunctors on $\mathbf{Set}$ preserve epimorphisms, it follows that the induced map $TZ \rightarrow T\Gamma_r$ is also epi. We have defined Tr to have the graph Γ_{Tr} which is the image of $T\Gamma_r$ in $TX \times TY$.



The result is that the induced map $TZ \rightarrow \Gamma_{Tr}$ is still epi and so $Tr = Tg \circ Tf^{-1}$ is independent of the factorization. Then to see that T is a pseudofunctor it is sufficient to show that $T(f^{-1}g) \subseteq Tf^{-1}Tg$, for then we have, after factoring $d_r^{-1}c_s = pq^{-1}$



$$\begin{aligned} T(rs) &= T(c_r d_r^{-1} \circ c_s d_s^{-1}) = T(c_r \circ (pq^{-1}) \circ d_s^{-1}) \\ &= T(c_r p) \circ T(q^{-1} d_s^{-1}) = Tc_r \circ Tp \circ T((d_s q)^{-1}) \\ &= Tc_r Tp \circ Tq^{-1} Td_s^{-1} = Tc_r \circ T(pq^{-1}) \circ Td_s^{-1} \\ &= Tc_r \circ T(d_r^{-1} c_s) \circ Td_s^{-1} \subseteq Tc_r Td_r^{-1} \circ Tc_s Td_s^{-1} \\ &= Tr \circ Ts \end{aligned}$$

Clearly equality holds if either r is a function or s is an inverse function. Now suppose that u and v are the pullback of f and g as in 1.1.

After we apply T , the diagram will still commute, but might fail to be a pullback. Thus

$$T(f^{-1}g) = T(uv^{-1}) = TuTv^{-1} \subseteq Tf^{-1}Tg$$

If $\alpha : T \Rightarrow S$ is a natural transformation of functors $\mathbf{Set} \rightarrow \mathbf{Set}$, it becomes pseudonatural when applied to the same functors on $\mathbf{Rel} \rightarrow \mathbf{Rel}$, which in this context means that

$$\alpha_Y Tr \leq Sr \alpha_X$$

Or, in diagram form

$$\begin{array}{ccc} TX & \xrightarrow{\alpha_X} & SX \\ \wr_{Tr} \downarrow & \lhd & \downarrow \wr_{Sr} \\ TY & \xrightarrow{\alpha_Y} & SY \end{array}$$

2 . Relational algebras.

We recall some of the definitions from the introduction to [B E69].

A triple $\mathbb{T} = (T, \eta, \mu)$ on C is an endofunctor $T : C \rightarrow C$ together with a natural transformation $\eta : T \rightarrow 1_X$ and $\mu : T^2 \rightarrow T$ satisfying the commutative diagrams in C^C

$$\begin{array}{ccc} T^3 & \xrightarrow{\mu_T} & T^2 \\ T\mu \downarrow & & \downarrow \mu \\ T^2 & \xrightarrow{\mu} & T \end{array} \quad \begin{array}{ccccc} T & \xrightarrow{\eta_T} & T^2 & \xleftarrow{T\eta} & T \\ & \searrow 1_T & \downarrow \mu & \swarrow 1_T & \\ & & T & & \end{array}$$

The category $C^{\mathbb{T}}$ of $\mathbb{T}$ -algebras, has as objects pairs (X, x) where X is an object and $x : TX \rightarrow X$ a morphism of C such that the following diagrams commute in C .

$$\begin{array}{ccc} TTX & \xrightarrow{\mu_X} & TX \\ Tx \downarrow & & \downarrow x \\ TX & \xrightarrow{x} & X \end{array} \quad \begin{array}{ccc} X & \xrightarrow{\eta_X} & TX \\ & \searrow 1_X & \downarrow x \\ & & X \end{array}$$

By a **relational $\mathbb{T}$ -prealgebra** we mean a pair (X, x) , where $x : TX \rightsquigarrow X$ is a relation.

A mapping $f : (X, x) \rightarrow (Y, y)$ of relational prealgebras is a function $f : X \rightarrow Y$ following the diagram in **Rel**

$$\begin{array}{ccc} TX & \xrightarrow{Tf} & TY \\ x \wr \downarrow & \lhd & \downarrow y \wr \\ X & \xrightarrow{f} & Y \end{array}$$

The category of relational prealgebra is denoted by $\mathbf{Set}^{P(\mathbb{T})}$. A relational prealgebra (X, x) is called a relational algebra if

$$\begin{array}{ccc} TTX & \xrightarrow{\mu_X} & TX \\ Tx \wr \downarrow & \lhd & \downarrow \wr \\ TX & \rightsquigarrow_x & X \end{array} \quad \begin{array}{ccc} X & \xrightarrow{\eta_X} & TX \\ & \searrow 1_X & \downarrow \wr x \\ & & X \end{array}$$

This definition is a precise generalization of algebra because any inclusion between actual maps in **Set** is an equality. We let $\mathbf{Set}^{R(\mathbb{T})} \subseteq \mathbf{Set}^{P(\mathbb{T})}$ denote the full subcategory consisting of relational algebras.

that $\mathbf{Set}^{\mathbb{T}} \subseteq \mathbf{Set}^{R(\mathbb{T})} \subseteq \mathbf{Set}^{P(\mathbb{T})}$ are full inclusions follows from the fact that an inclusion among functions is an equality. We will show that each is a coreflective inclusions (by which we mean it has a left adjoint). First we show that limits and colimits are easily computed in $\mathbf{Set}^{P(\mathbb{T})}$.

Proposition 2.1. *$\mathbf{Set}^{P(\mathbb{T})}$ is complete and cocomplete.*

Proof. Limits (=projective limits) are computed in $\mathbf{Set}^{P(\mathbb{T})}$ exactly as in $\mathbf{Set}^{\mathbb{T}}$. For example if $(X_i, x_i)_{i \in I}$ is a family of relational prealgebras, their product in $\mathbf{Set}^{P(\mathbb{T})}$ is $(\prod_I X_i, \prod_I x_i \circ a)$ where a satisfies for all $i \in I$

$$\begin{array}{ccc} T(\prod_I X_i) & \xrightarrow{a} & \prod_I T(X_i) \\ & \searrow T(\pi_i) & \downarrow \pi_i \\ & & T(X_i) \end{array}$$

(where π_i is the i 'th coordinate projection.)

Equalizers are computed similarly. The same holds for relational algebras.

As for colimits, first consider the same family $(X_i, x_i)_{i \in I}$. Let $b : \coprod_I T(X_i) \rightarrow T(\coprod_I X_i)$ be determined by

$$\begin{array}{ccc} T(\coprod_I X_i) & \xleftarrow{b} & \coprod_I T(X_i) \\ & \nwarrow T(\iota_i) & \uparrow \iota_i^T \\ & & T(X_i) \end{array}$$

(where ι_i is the i 'th coordinate injection; We distinguish between $\iota_i : X_i \rightarrow \coprod_I X_i$ and $\iota_i^T : TX_i \rightarrow \coprod_I TX_i$). Then I claim that $(\coprod_I X_i, \coprod_I x_i \circ b^{-1})$ is the coproduct. First we must show that the $\iota_i : X_i \rightarrow \coprod_I X_i$ are morphisms, i.e. that

$$\begin{array}{ccc} T(X_i) & \xrightarrow{T(\iota_i)} & T(\coprod_I X_i) \xrightarrow{\sim b^{-1}} \coprod_I T(X_i) \\ x_i \downarrow & \leq & \downarrow \coprod_I x_i \\ X_i & \xrightarrow{\iota_i} & \coprod_I X_i \end{array}$$

Now we can build (using proposition 1.2)

$$\begin{array}{ccccc} & & T(\coprod_I X_i) & & \\ & \nearrow T\iota_i & \uparrow b & \nwarrow \sim b^{-1} & \\ TX_i & \xrightarrow{\iota_i^T} & \coprod_I TX_i & \xrightarrow{1_{\coprod_I TX_i}} & \coprod_I TX_i \end{array}$$

In other words, $\iota_i^T \leq b^{-1}T\iota_i$

Thus when applying to the coproduct diagram, we obtain

$$\begin{array}{ccccc} X_i & \xleftarrow{x_i} & TX_i & \xrightarrow{T\iota_i} & T(\coprod_I X_i) \\ \iota_i \downarrow & & \downarrow \iota_i^T & \leq \nwarrow \sim b^{-1} & \\ \coprod_I X_i & \xleftarrow{\coprod_I x_i} & \coprod_I TX_i & & \end{array}$$

Which (ignoring ι_i^T) is exactly the diagram we chased.

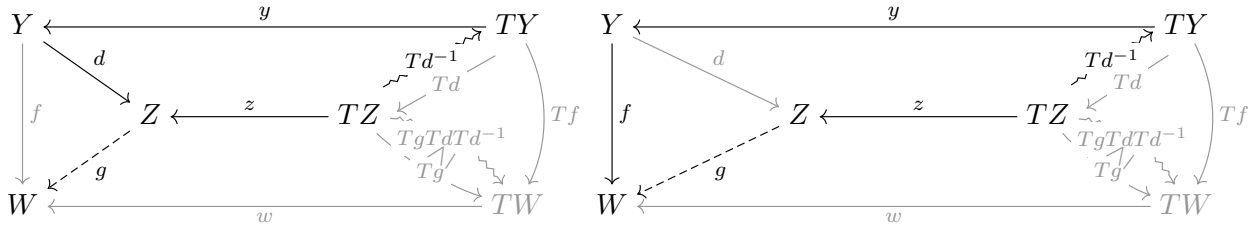
Now, if $f_i : (X_i, x_i) \rightarrow (Y, y)$ is given for each $i \in I$, then the f_i 's extend to a unique $f : \coprod_I X_i \rightarrow Y$ such that $f \circ \iota_i = f_i$. In diagram form

$$\begin{array}{ccc} X_i & \xrightarrow{f_i} & Y \\ \iota_i \downarrow & \nearrow \exists! f & \\ \coprod_I X_i & & \end{array}$$

Now following this diagram

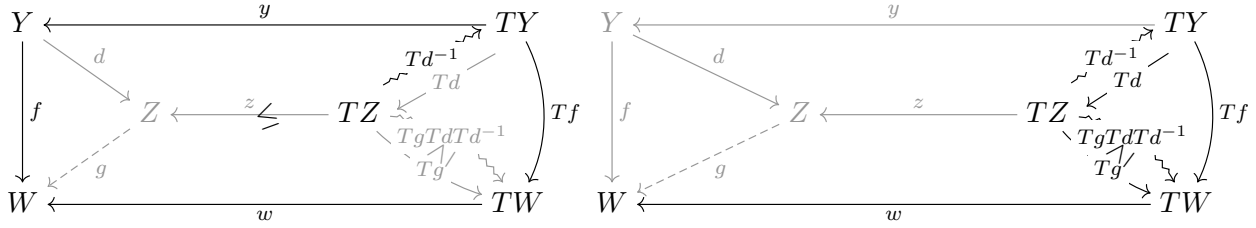
$$\begin{array}{ccccccc} \coprod_I TX_i & \xleftarrow{\iota_i^T} & TX_i & \xrightarrow{\iota_i^T} & \coprod_I TX_i & & \\ & \searrow x_i & \downarrow Tf_i & \searrow T\iota_i & \downarrow b & & \\ \coprod_I x_i & & X_i & \leq & TY & \xleftarrow{Tf} & T\coprod_I X_i \\ & \nwarrow \iota_i & \searrow f_i & & \downarrow y & \nearrow & \\ & & \coprod_I X_i & \xrightarrow{f} & Y & & \end{array}$$

$$f \circ \coprod_I x_i \circ \iota_i^T = f \circ \iota_i \circ x_i = f_i \circ x_i \leq y \circ Tf_i = y \circ Tf \circ T\iota_i = y \circ Tf \circ b \circ \iota_i^T$$



$$g \circ z \equiv g \circ d \circ y \circ Td^{-1}$$

$$g \circ z \equiv f \circ y \circ Td^{-1}$$



$$f \circ y \circ Td^{-1} \leq w \circ Tf \circ Td^{-1}$$

$$= w \circ Tg \circ Td \circ Td^{-1} \leq w \circ Tg$$

□

Remark. Note that the limits and colimits preserve the underlying sets. This could have been predicted from the fact that the underlying set functor $\mathbf{Set}^{P(\mathbb{T})} \rightarrow \mathbf{Set}$ has both adjoints, namely

$$X \mapsto (X, \eta_X^{-1}) \text{ and } X \mapsto (X, 1_X \times 1_{TX})$$

being left and right adjoint, respectively. Each of these is a relational algebra so the same remarks apply to $\mathbf{Set}^{R(\mathbb{T})} \rightarrow \mathbf{Set}$

Proposition 2.2. (X, x) is a relational algebra if and only if $x \geq \eta_X^{-1}$ and $x \circ Tx \circ \mu_X^{-1} \leq x$

$$\begin{array}{ccc} TX & & TX \xleftarrow{Tx} TTX \\ \eta_X^{-1} \downarrow \leq & & x \downarrow \quad \vee \quad \uparrow \mu_X^{-1} \\ X & & X \xleftarrow{x} TX \end{array}$$

Proof. If (X, x) is a relational algebra then $1_X \subseteq x\eta_X$ so

$$\begin{array}{ccc} TX & \xrightarrow{\eta_X^{-1}} & X \\ \eta_X^{-1} \downarrow & \nearrow 1_X & \downarrow \eta_X \\ X & \xleftarrow{x} & TX \end{array} \quad \text{thus} \quad \begin{array}{ccc} TX & \equiv & TX & \equiv & TX \\ \eta_X^{-1} \downarrow & \leq & x\eta_X \eta_X^{-1} & \leq & \downarrow x \\ X & \equiv & X & \equiv & X \end{array}$$

To see the converse we suppose that $x \geq \eta_X^{-1}$. Then

$$\begin{array}{ccc} TX & \xleftarrow{\eta_X} & X \\ x \downarrow \geq \eta_X^{-1} & & \uparrow 1_X \end{array}$$

Similarly, if $x \circ Tx \circ \mu_X^{-1} \leq x$, then

$$x \circ Tx \leq x \circ Tx \circ \mu_X^{-1} \mu_X \leq x \circ \mu_X$$

In diagram form

$$\begin{array}{ccc}
 TTX & & \\
 \downarrow 1_{TTX} & \searrow \mu_X & \\
 TTX & \xleftarrow{\mu_X^{-1}} & TX \\
 \downarrow Tx & \searrow \mu_X^{-1} & \downarrow \\
 TX & \xrightarrow{x} & X
 \end{array}$$

□

Proposition 2.3. *The obvious functor $\mathbf{Set}^{R(\mathbb{T})} \rightarrow \mathbf{Set}^{P(\mathbb{T})}$ has a left adjoint.*

Proof. Fix X and let $\eta = \eta_X$ and $\mu = \mu_X$. If (X, x) is a relational prealgebra, we define an ordinal sequence of reations $x_n : TX \rightarrow X$ as follows:

- Let $x_0 = x \vee \eta^{-1}$
- $x_n = \bigvee_{m < n} x_m$ for n a limit ordinal,
- $x_n = x_{n-1} \circ Tx_{n-1} \circ \mu^{-1}$ otherwise.

Clearly $x_0 \geq x$ and $x_0 \geq \eta^{-1}$. Assuming $x_m \geq x_k$ for all $k < m < n$, we have $x_n \geq x_k$ for all $k < n$ if n is a limit ordinal, otherwise

$$x_n = x_{n-1}Tx_{n-1}\mu^{-1} \geq x_{n-1}T\eta^{-1}\mu^{-1} = x_{n-1}(\mu T\eta)^{-1} = x_{n-1}$$

Now since $TX \times X$ is a set, this ascending chain of subsets represented by the graphs of these x_n must eventually terminate, at which point we call $\bar{x} := x_n = x_{n-1}$, and so by 2.2, (Tx, x) is a relational algebra. If $f : (X, x) \rightarrow (Y, y)$ is a morphism of (X, x) into the relational algebra (Y, y) , then $f \circ x \leq y \circ Tf$. Also considering the diagram

$$\begin{array}{ccccc}
 & & TX & & \\
 & \nearrow \eta_X & & \searrow Tf & \\
 X & & & & TY \\
 & \searrow f & & \nearrow \eta_Y & \\
 & & Y & &
 \end{array}$$

as a diagram like 1.1, we have

$$f \circ \eta_X^{-1} \leq \eta_Y^{-1} \circ Tf \leq y \circ Tf$$

so that $f \circ x_0 \leq y \circ Tf$.

Similarly if n is a limit ordinal and $f \circ x_m \leq y \circ Tf$ for all $m < n$, then certainly $f \circ x_n \leq y \circ Tf$.

Finally, if $f \circ x_{n-1} \leq y \circ Tf$, then

$$f \circ x_n = f \circ x_{n-1} \circ Tx_{n-1} \circ \mu^{-1} \leq y \circ T(f \circ x_{n-1})\mu^{-1}$$

(since f is a function)

$$\begin{array}{ccccccc}
 TX & \xlongequal{\quad} & TX & & TX & & TX \\
 \downarrow x_n & \searrow \mu^{-1} & \downarrow & \searrow \mu & \downarrow & \searrow \mu^{-1} & \downarrow \\
 & TTX & \xlongequal{\quad} & TTX & \xlongequal{\quad} & TTX & \xlongequal{\quad} TTX \\
 & \downarrow Tx_{n-1} & & \downarrow & & \downarrow T^2f & \\
 & TX & \xlongequal{\quad} & TX & \xlongequal{\quad} & TTY & \xlongequal{\quad} TTY \\
 & \downarrow x_{n-1} & & \downarrow & & \downarrow Ty & \\
 X & \xlongequal{\quad} & X & \xrightarrow{\quad} & X & \xrightarrow{\quad} & X \\
 \downarrow f & & \downarrow f & & \downarrow f & & \downarrow f \\
 Y & \xrightarrow{\quad} & Y & \xrightarrow{\quad} & TY & \xrightarrow{\quad} & TY \\
 & & y & & & &
 \end{array}$$

We get that in the end $f x_n \leq y Tf$. Hence $f \circ \bar{x} \leq y \circ Tf$ from which the result easily follows.

□

Proposition 2.4. *The obvious functor $\mathbf{Set} \rightarrow \mathbf{Set}^{R(\mathbb{T})}$ has a left adjoint*

Proof. This proof differs from the preceding in two respects. First the adjoint does not preserve the underlying set (by example). Second, the proof is nonconstructive, being an application of the adjoint functor theorem. It is clear from the description of limits in $\mathbf{Set}^{P(\mathbb{T})}$ that the functor $\mathbf{Set}^{\mathbb{T}} \rightarrow \mathbf{Set}^{P(\mathbb{T})}$ preserves them. Also $\mathbf{Set}^{R(\mathbb{T})}$ being a coreflective subcategory of $\mathbf{Set}^{P(\mathbb{T})}$ computes its limits in that category, so that $\mathbf{Set}^{\mathbb{T}} \rightarrow \mathbf{Set}^{R(\mathbb{T})}$ preserves limits. So it is only necessary to find a solution set. Given (X, x) , consider all algebras of the form $(TX, \mu_X)/\sim$, for which the composite

$$X \xrightarrow{\eta_X} TX \longrightarrow TX/\sim$$

is a mapping of relational algebras. This is clearly small.

If $f : (X, x) \rightarrow (Y, y)$ is a mapping of X into a $\mathbb{T}$ -algebra, then $yTf : TX \rightarrow Y$ is a function and

$$\begin{array}{ccccc} TTX & \xrightarrow{TTf} & TTY & & \\ \mu_X \downarrow & \searrow T(yTf) & \downarrow \mu_Y & & \\ TX & \xrightarrow{Tf} & TY & \xrightarrow{y} & Y \end{array}$$

$$yTf\mu_X = y \circ Y(yTf)$$

thus $yTf : (TX, \mu_X) \rightarrow (Y, y)$ is a map of algebras. As usual it factors as

$$(TX, \mu_X) \xrightarrow{b} (Y_0, y_0) \xrightarrow{a} (Y, y)$$

Where (Y_0, y_0) is a factor algebra of (TX, μ_X) and a is monic. Then

$$\begin{array}{ccccc} X & \xrightarrow{\eta_X} & TX & \xrightarrow{b} & Y_0 \\ \downarrow f & & \downarrow Tf & & \downarrow a \\ Y & \xrightarrow{\eta_Y} & TY & \xrightarrow{y} & Y \\ & & \uparrow \eta_Y & & \\ & & 1_Y & & \end{array}$$

In other words $a \circ b \circ \eta_X = f$

$$\begin{array}{ccccccc} & & TX & \xrightarrow{x} & X & \xrightarrow{\eta_X} & TX & \xrightarrow{b} & Y_0 \\ & \swarrow T(b\eta_X) & \downarrow Tf & \geq & \downarrow f & & & & \downarrow a \\ TY_0 & \xrightarrow{Ta} & TY & \xrightarrow{y} & Y & \xlongequal{\quad} & Y & & \\ \downarrow y_0 & & & & & & \parallel & & \\ Y_0 & \xrightarrow{\quad} & & & & & Y & & \end{array}$$

and a is monic so $b\eta_X \circ x \leq y_0 \circ T(b\eta_X)$. Thus $b\eta_X : (X, x) \rightarrow (Y_0, y_0)$ factors f and the codomain is a member of the given set. We note in passing that this gives a new proof of the cocompleteness of $\mathbf{Set}^{\mathbb{T}}$ $\square$

3 . Topological spaces

theorem 3.1. *There's a natural equivalence J between the category **Top** of topological spaces and continuous maps and the category $\mathbf{Set}^{R(\mathbb{B})}$ where $\mathbb{B} = (\beta, \eta, \mu)$ is the compact Hausdorff space triple (ultrafilter triple).*

The proof is given by a series of propositions. We let $\mathcal{U}$ denote an ultrafilter on X .

Proposition 3.2. *If C is a topological space, $X = UC$ is its underlying set and $x : \beta X \rightarrow X$ is given by taking $x(\mathcal{U})$ to be the set of limits of $\mathcal{U}$, then*

$$x\eta \geq 1_X \text{ and } x\beta x \leq x\mu$$

Proof. The first statement is trivial, being merely the statement that if $\mathcal{P}$ is the principal ultrafilter at the point $p \in X$, then $p \in x(\mathcal{P})$, which is certainly true in any topological space. We can factor x as

$$\beta X \xleftarrow{j} DX \xleftarrow{d'x} \Gamma_x \xrightarrow{cx} X$$

where j is mono, and $d'x$ and cx are epi (the latter because of principal ultrafilters). β preserves both monos and epis so that

$$\beta^2 X \xleftarrow{\beta j} \beta DX \xleftarrow{\beta d'x} \beta \Gamma_x \xrightarrow{\beta cx} \beta X$$

is the same type of decomposition. The mapping βj consists of taking an ultrafilter on DX and using it as a filter base on βX . The filter generated is an ultrafilter. This if $\mathcal{A} \in \beta^2 X$, $\beta x(\mathcal{A}) \neq \emptyset$ if and only if given $A \in \mathcal{A}$, $DA = A \wedge DX \in \mathcal{A}$ also.

Now suppose that $\mathcal{A} \in \beta^2 X$, $\beta x(\mathcal{A}) = \mathcal{U}$ and $x(\mathcal{U}) = p$. Suppose moreover that $\mu(\mathcal{A}) = \mathcal{V}$. Then we must show that $x(\mathcal{V}) = p$ as well. This means showing that every open neighbourhood U of p is in $\mathcal{V}$ or, from the definition of μ , that $\{\mathcal{W} \mid U \in \mathcal{W}\} \in \mathcal{A}$. Now U is open, which means that $x(\mathcal{W}) \wedge U \neq \emptyset$.

Suppose we had $A = \{\mathcal{W} \mid U \notin \mathcal{W}\} \in \mathcal{A}$. Now there is an ultrafilter $\mathcal{B}$ on $\beta \Gamma_x$ whose projections are $\mathcal{A}$ and $\mathcal{U}$ and there would have to be a $B \in \mathcal{B}$ whose projections were A on one hand and some U_1 on the other.

If $(\mathcal{W}, q) \in B$, then $U \notin \mathcal{W}$, but $q \in x(\mathcal{W})$. As noted above, $x(\mathcal{W}) \wedge U = \emptyset$, and so $q \notin U$.

Therefore $U \cap U_1 = \emptyset$, which is a contradiction.

Hence $A \notin \mathcal{A}$, and since $\mathcal{A}$ is an ultrafilter, its complement $\{\mathcal{W} \mid U \in \mathcal{W}\} \in \mathcal{A}$. This completes the proof. $\square$

Proposition 3.3. *If C and D are topological spaces with underlying sets X and Y and convergence mappings x and y respectively, then $f : X \rightarrow Y$ underlies a continuous mapping if and only if $fx \circ y \circ \beta f$*

Proof. This is nothing but a translation of a well known theorem which states that f is continuous if and only if whenever $\mathcal{U}$ is an ultrafilter on X and $\mathcal{U}$ converges to p , then $\beta f(\mathcal{U})$ converges to $f(p)$ (see [RE60]). $\square$

Corollary 3.4. *There is a natural $J : \mathbf{Top} \rightarrow \mathbf{Set}^{R(\mathbb{B})}$ which is full and faithful*

Proof. Of course J is defined by $JC = (UC, \text{conv})$ as above and the preceding two propositions state that J is a well-defined functor and is full. It is also clearly faithful since U is. $\square$

Now suppose (X, x) is a relational algebra. We must show it is J of something. if $e : U \hookrightarrow X$ is a subset inclusion, we define a new subset

$$\bar{e} : \bar{U} \hookrightarrow X$$

by the following diagram

$$\begin{array}{ccccc} \beta X & \xleftarrow{dx} & \Gamma_x & \xrightarrow{cx} & X \\ \beta e \uparrow & & g \uparrow & & \uparrow \bar{e} \\ \beta U & \xleftarrow{dU} & \Gamma_U & \xrightarrow{cU} & \bar{U} \end{array}$$

Fig. 3.5

in which Γ_U is the pullback of βe and dx and $\bar{e} \circ cU$ is the mono/epi factorization of $cx \circ g$. In words, $\bar{U}$ is the set of “limits” of all ultrafilters containing U .

Proposition 3.6. $U \mapsto \bar{U}$ is a closure operator.

Proof. We must show

- a) $\bar{\emptyset} = \emptyset$,
- b) $U \leq \bar{U}$,
- c) $\overline{U_1 \vee U_2} = \bar{U}_1 \vee \bar{U}_2$, and
- d) $\bar{U} = \bar{\bar{U}}$.

Let us prove them:

- a) is trivial since $\beta\emptyset = \emptyset$ and any pullback along $\emptyset$ is empty.
- b) is also easy or $p \in U$; the principal ultrafilter $\mathcal{P}$ contains U and converges to p . This implies that $\bar{U}_1 \vee \bar{U}_2 \geq \bar{U}_1 \vee \bar{U}_2$, so showing
- c) requires only the reverse inclusion. But if an ultrafilter $\mathcal{U}$ contains $U_1 \vee U_2$, then it must contain one of them, and so we get the reverse inclusion. For
- d) it is easily seen to be sufficient to find a relation $t : \bar{\bar{U}} \rightsquigarrow U$ such that $\bar{e} \leq \bar{e}t$, for then

$$\text{image}(\bar{e}) \leq \text{image}(\bar{e}t) \leq \text{image}(\bar{e})$$

proving the result. We need t such that

$$\begin{array}{ccc} \bar{\bar{U}} & \xrightarrow{t} & \bar{U} \\ \bar{e} \downarrow & \searrow \bar{e} & \\ X & & \end{array}$$

We start by noticing that $\bar{e} = \bar{e}c\bar{U}c\bar{U}^{-1}$

$$\begin{array}{ccccc} \bar{\bar{U}} & \xrightarrow{c\bar{U}^{-1}} & \Gamma_{\bar{U}} & \xrightarrow{c\bar{U}} & \bar{\bar{U}} \\ \bar{e} \downarrow & & & & \downarrow \bar{e} \\ X & \xlongequal{\quad} & & & X \end{array}$$

and by 3.5 $\bar{e}c\bar{U} = cx\bar{g}$ thus

$$\begin{array}{ccccccc} & & & & \beta X & & \\ & & & & \nearrow dx & \downarrow \beta dx^{-1} & \\ \bar{\bar{U}} & \xrightarrow{c\bar{U}^{-1}} & \Gamma_{\bar{U}} & \xrightarrow{g} & \Gamma_x & \xrightarrow{1_{\Gamma_x}} & \Gamma_x \\ \bar{e} \downarrow & & & & & & \downarrow cx \\ X & \xlongequal{\quad} & & & & & X \end{array}$$

now we know $cx \circ dx^{-1} = x$ and $dx \circ \bar{g} = \beta\bar{e} \circ d\bar{U}$, thus

$$\begin{array}{ccccccc} & & \Gamma_x & & & & \beta\Gamma_U = \beta\Gamma_U \\ & & \nearrow g & \searrow dx & & & \uparrow \beta cU^{-1} \quad \downarrow \beta cU \\ \bar{\bar{U}} & \xrightarrow{c\bar{U}^{-1}} & \Gamma_{\bar{U}} & \xrightarrow{d\bar{U}} & \beta\bar{U} & \xrightarrow{1_{\beta\bar{U}}} & \beta\bar{U} & \xrightarrow{\beta\bar{e}} & \beta X \\ \bar{e} \downarrow & & & & \downarrow \beta\bar{e} & & \downarrow \beta\bar{e} & & \downarrow x \\ X & \xlongequal{\quad} & & & X & \xlongequal{\quad} & X & \xlongequal{\quad} & X \end{array}$$

and finally $cx \circ g = \bar{e} \circ cU$, thus

$$\begin{array}{ccccccc}
 \Gamma_{\bar{U}} & \xrightarrow{d\bar{U}} & \beta\bar{U} & \xrightarrow{\beta cU^{-1}} & \beta\Gamma_U & \xrightarrow{\beta dU} & \beta^2 U & \xrightarrow{\mu_U} & \beta U & \xrightarrow{dU^{-1}} & \Gamma_U \\
 \uparrow c\bar{U}^{-1} & & & & & & & & & & \nwarrow cU \\
 \bar{\bar{U}} & & & & & & & & & & \Gamma_x \\
 \downarrow \bar{e} & & & & & & & & & & \swarrow cx \\
 X & \xlongequal{\hspace{1.5cm}} & & & & & & & & & X
 \end{array}$$

Thus we can define²

$$t = cU dU^{-1} \mu_U dU \beta cU^{-1} d\bar{U} c\bar{U}^{-1}$$

giving us the inequality we required.

In words, take a point of $\bar{\bar{U}}$ and find an ultrafilter on $\bar{U}$ converging to this. Represent that in turn by an ultrafilter on βU . Converge that under μ to an ultrafilter on U and converge that under x .

□

It is well known that such a closure operator induces a unique topology on X . There is now only one thing left to complete the proof of theorem 3.1.

Proposition 3.7. *Suppose $(X, x) \in \mathbf{Set}^{R(\mathbb{B})}$ and the closure is defined as above. Then for any $\mathcal{U} \in \beta X$, $p \in x(\mathcal{U})$ if and only if $p \in \bigcap \{\bar{U} \mid U \in \mathcal{U}\}$*

Remark. *Since this latter is the formula for ultrafilter convergence under a closure operator, this means that x is convergence (as described in the introduction) under that topology.*

Proof. One way is clear: if $p \in x(\mathcal{U})$ and $U \in \mathcal{U}$, $\mathcal{U}$ is an ultrafilter containing U and p is one of its limits under x so $p \in \bar{U}$.

To go the other way, we suppose that $U \in \mathcal{U}$ converges to $p \in \bar{U}$. We want to show that $p \in x(\mathcal{U})$. The idea of the proof is to find an $\mathcal{A} \in \beta^2 X$ with $\mu(\mathcal{A}) = \mathcal{U}$, while $\beta x(\mathcal{A}) = \mathcal{P}$, the principal ultrafilter at p . Then $p \in x(\mathcal{P}) \leq x\mu(\mathcal{A}) = x(\mathcal{U})$.

We begin by constructing an ultrafilter $\mathcal{B}$ on Γ_x : Let

$$B = \{(\mathcal{V}, q) \mid p \in x(\mathcal{V})\},$$

and for each $U \in \mathcal{U}$ let

$$B_U = \{(\mathcal{V}, q) \mid U \in \mathcal{V}, q \in x(\mathcal{V})\}.$$

Then for $U_1, U_2 \in \mathcal{U}$

$$B_{U_1} \wedge B_{U_2} \geq B_{U_1 \wedge U_2} \neq \emptyset$$

since $U_1 \wedge U_2 \neq \emptyset$. Moreover, since $p \in \bar{U}$ for each $U \in \mathcal{U}$, we can find a $\mathcal{V}$ such that $U \in \mathcal{V}$ and $p \in x(\mathcal{V})$. Then $(\mathcal{V}, p) \in B \wedge B_U$ thus B together with $\{B_U\}$ generates a proper filter which is contained in an ultrafilter $\mathcal{B}$ on Γ_x .

Clearly one projection is $\mathcal{P}$ since B projects to $\{p\}$. The other projection is $\mathcal{A}$ and it is clear that for all $U \in \mathcal{U}$,

$$\{\mathcal{V} \mid U \in \mathcal{V}\} \geq dx(B_U)$$

and thus is in $\mathcal{A}$.

Hence $U \in \mu(\mathcal{A})$ for all $U \in \mathcal{U}$, and so $\mathcal{U} = \mu(\mathcal{A})$

□

² Note: This proof has been changed from the original; I believe that providing t as a composition of 7 relations is not very intuitive and it is necessary to show where this comes from.

4 . Other examples

In this section we consider a few other examples of relational algebras although in no case are the results as striking as the preceding. The simplest example is the identity triple $\mathbb{I} = (1, 1, 1)$.

Here the laws simply reduce to $x : X \rightsquigarrow X$ subject to

$$x \circ x \leq x \text{ and } x \geq 1_X$$

Thus an algebra for this triple is a set with a transitive, reflexive relation, usually called a preordered set. It is simple to check that mappings are exactly the order preserving functions. It is perhaps interesting to note in this connection that the algebras are a coreflective (hence tripleable) subcategory of preordered sets.

The next example is the triple $\mathbb{T} = (T, \eta, \mu)$, where $TX = X + 1$, where $+$ denotes the coproduct. The category of algebras is the category of pointed sets. A relational algebra is a set X with a relation $x : X + 1 \rightsquigarrow X$. Now $X + 1$ is a coproduct and so x is determined by its restrictions $x|_X$ and $x|_1$.

The first restriction is a preorder, as before, and the second is just a subset $X_0 \leq X$. The unitary law is automatic (as soon as $x|_X$ is a preorder) while the associative law requires that X_0 be a ray. That is if $p \in X_0, q \leq_X p$ (or $q \in x(p)$), then $q \in X_0$ also. Thus the algebras are pairs $(X, X_0 \leq X)$ where X is a preordered set and X_0 is a ray.

Morphisms are maps of pairs which preserve the order as well.

Similar considerations apply to the category of relational models of the triple where TX is the set of ultrafilters on $X + 1$

The algebras are pairs (X, X_0) where X is a topological space and $X_0 \leq X$ is a closed subspace. Mappings are continuous maps if pairs.

No other interesting examples are known to us. It seems fairly clear that **Cat**, for example, is a full subcategory of the relational models of the theory of monoids (however the nullary operation 1 becomes the set of objects of a category!) but just being a full subcategory is not very informative. (E.g. the recent work of the “Čech school” seems to show that practically everything is fully embedded in the category of semigroups).

References

- [R E60] W.H. Gottschalk R. Ellis. “Homomorphisms of transformation groups”. **in** *Trans. Amer. Math. Soc.*: 94.258-271 (1960).
- [Lan61] S. Mac Lane. “An Algebra of additive Relations”. **in** *Proc. NAS-USA*: 47.1043-1051 (1961).
- [B E69] ed. B. Eckman. “Seminar on Triples and Algebraic Homology Theory”. **in** *Lecture Notes in Mathematics*, no. 80: **byeditor**Springer Verlag. 1969.
- [Bar69] Michael Barr. “Relational algebras”. **in** *Reports of the Midwest Category Seminar IV*: Berlin, Heidelberg: Springer Berlin Heidelberg, 1969, **pages** 39–55. ISBN: 978-3-540-36292-0.
- [Man69] E. Manes. “A triple theoretic construction of compact algebras”. **in** *Lecture Notes in Mathematics*, no. 80: **byeditor**Springer Verlag. 1969, **pages** 91–118.